

ASSIGNMENT 2

- **SESSION 20.**

1.Create a table comparing Agile and Waterfall models across 5 factors relevant to software testing, such as requirements, feedback cycle, tester involvement, documentation, and flexibility.

ANS.

Factor	Agile	Waterfall
Requirements	Requirements can change throughout the project and are refined every sprint.	Requirements are defined at the beginning and changes are difficult to accommodate.
Feedback Cycle	Continuous feedback from customers and stakeholders after each sprint.	Feedback is usually received after the complete product or a major phase is finished.
Tester Involvement	Testers are involved from the beginning and work alongside developers.	Testers are mainly involved after the development phase is completed.
Documentation	Focuses on lightweight documentation with emphasis on working software.	Requires detailed documentation for each phase before moving to the next.
Flexibility	Highly flexible; changes can be incorporated quickly.	Less flexible; changes often require revisiting completed phases.

2. Pick any app you use daily (like Instagram, Zomato, or Paytm) and describe how the testing process would look different if the team used Agile instead of Waterfall. Mention at least 3 specific testing activities or practices that would change.

Ans.

If **Zomato** were developed using **Agile** instead of **Waterfall**, the testing process would be more continuous and collaborative.

Testing Activity	Waterfall	Agile
Test Case Creation	Created after requirements and development are complete.	Created alongside user stories during each sprint.
Testing Frequency	Performed after the entire module is developed.	Performed continuously during every sprint.
Bug Fixing	Bugs are fixed after the testing phase, often near project completion.	Bugs are fixed immediately within the same sprint whenever possible.

Specific Agile Testing Practices :-

1. Regression testing is executed after every sprint.
2. Daily communication between developers and testers helps identify and resolve defects quickly.

3. Draw a simple workflow diagram showing the main roles (such as Scrum Master, Product Owner, Development Team, Tester) and ceremonies (like Sprint Planning, Daily Standup, Sprint Review) in a SCRUM process.

Ans.

SCRUM Workflow Diagram :-

Main Roles :-

- **Product Owner** – Defines and prioritizes product requirements.
- **Scrum Master** – Facilitates the Scrum process and removes obstacles.
- **Development Team** – Designs, develops, and fixes features.
- **Tester (QA)** – Designs test cases, executes tests, reports defects, and verifies fixes.

Main Ceremonies :-

- Sprint Planning
- Sprint Review
- Sprint Retrospective

4. For a food delivery app (like Swiggy), map out which testing activities should happen during each phase of a typical Agile sprint.

Ans.

Sprint Phase	Testing Activities
Backlog Grooming	Review user stories, clarify acceptance criteria, identify test scenarios, estimate testing effort.
Sprint Planning	Analyze sprint stories, prepare test strategy, design test cases, identify test data requirements.
Development	Review code changes, perform unit testing (developers), prepare test environment, begin functional testing as features become available.
Test Execution	Execute functional, integration, and regression tests; log defects; verify bug fixes; update test results.
Sprint Review	Demonstrate completed features, validate acceptance criteria, collect stakeholder feedback, perform final verification.
Sprint Retrospective	Discuss testing challenges, analyze escaped defects, identify process improvements, and define action items for the next sprint.

Typical Testing Flow During an Agile Sprint :-

1. User stories are refined during **Backlog Grooming**.
2. Test cases are designed during **Sprint Planning**.
3. Testing begins as soon as features are developed.
4. Defects are reported and verified continuously throughout the sprint.
5. Regression testing is performed before the sprint ends.
6. During the **Sprint Review**, completed features are validated against acceptance criteria.
7. Lessons learned are discussed in the **Sprint Retrospective** to improve future testing.